Retrieval-Augmented Generation (RAG) Test Document

Introduction

Retrieval-Augmented Generation (RAG) combines information retrieval with language generation. Instead of relying only on model parameters, a RAG system searches a knowledge base, retrieves relevant passages, and uses them to answer questions. This document intentionally repeats key concepts with varied wording so that both Multi-Query Retrieval (MQR) and contextual retrieval systems can be tested.

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Core Workflow

A typical pipeline includes document ingestion, text extraction, chunking, embedding generation, vector indexing, retrieval, reranking, prompt construction, and final answer generation. Chunk overlap helps preserve context across boundaries.

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Chunking

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Embeddings

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Vector Databases

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Multi-Query Retrieval

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Contextual Retrieval

Contextual retrieval expands or enriches retrieved chunks with nearby information. If a paragraph references 'this method', surrounding paragraphs may be included so the model understands what 'this method' refers to.

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Evaluation

Useful metrics include Precision@K, Recall@K, Mean Reciprocal Rank, answer faithfulness, and latency. High retrieval quality generally improves grounded responses.

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Example Scenario

A company stores employee manuals, HR policies, technical documentation, and troubleshooting guides. Employees ask natural language questions, and the assistant retrieves only the relevant documents before answering.

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Conclusion

An effective RAG system balances chunk size, embedding quality, retrieval strategy, reranking, and prompt design. Testing with paraphrased queries, ambiguous wording, and context-dependent questions helps measure real-world performance.

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